

KRASNOUFIMSK, Russian Federation

WMO: 284340

Lat: **56.650N**Lon: **57.783E**Elev: **206**StdP: **98.87**Time Zone: **5.00 (E05)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.0	-31.7	-37.7	0.1	-34.8	-34.3	0.2	-31.6	6.3	-6.8	5.4	-6.7	0.2	340	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	29.7	19.7	27.9	19.1	26.1	18.0	21.0	28.0	20.0	26.2	19.0	24.5	2.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.6	13.8	24.0	17.7	13.0	23.0	16.8	12.3	21.9	61.5	28.1	58.2	26.5	54.9	24.5	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.9	4.3	DB	-39.4	32.2	2.8	1.8	-41.5	33.6	-43.1	34.6	-44.7	35.6	-46.8	37.0	(4)
(5)			WB	-39.5	22.2	2.8	1.3	-41.5	23.2	-43.2	23.9	-44.7	24.7	-46.8	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-15.4	-13.8	-5.7	3.4	11.1	16.0	17.9	15.6	10.0	3.2	-5.8	-12.5	(6)	
	DBStd	13.24	8.64	8.17	5.59	5.17	4.88	4.19	3.73	3.96	4.22	4.55	7.51	8.62	(7)	
	HDD10.0	3636	786	667	486	204	47	5	1	3	51	214	475	698	(8)	
	HDD18.3	6016	1044	901	744	447	229	92	54	103	251	469	725	956	(9)	
	CDD10.0	753	0	0	0	7	80	187	247	178	51	3	0	0	(10)	
	CDD18.3	91	0	0	0	0	4	23	42	20	1	0	0	0	(11)	
	CDH23.3	1129	0	0	0	3	87	299	496	224	19	0	0	0	(12)	
	CDH26.7	294	0	0	0	0	11	77	138	66	2	0	0	0	(13)	
(14) Wind	WSAvg	1.7	1.5	1.7	2.0	2.0	1.9	1.5	1.2	1.2	1.4	1.9	1.8	1.7	(14)	
(15) Precipitation	PrecAvg	531	32	23	19	30	40	65	77	64	55	48	42	33	(15)	
	PrecMax	812	75	85	59	84	78	129	180	124	122	118	89	67	(16)	
	PrecMin	309	3	0	0	3	3	17	19	22	10	9	11	3	(17)	
	PrecStd	94	15	17	13	21	18	29	44	26	28	22	18	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.5	2.1	9.5	22.6	28.0	31.4	31.9	31.6	26.4	18.3	8.7	1.8	(19)	
		MCWB	0.4	0.6	4.5	12.2	15.6	19.7	21.0	20.5	15.9	11.4	6.4	0.8	(20)	
	2%	DB	-0.1	0.7	5.5	18.8	25.8	28.7	29.9	28.4	22.5	15.0	5.8	0.9	(21)	
		MCWB	-1.0	-0.5	2.2	10.4	15.2	19.1	20.2	19.8	15.0	9.6	4.4	0.0	(22)	
	5%	DB	-1.5	-0.6	3.4	15.2	23.5	26.8	28.2	25.7	19.6	12.4	3.8	0.0	(23)	
		MCWB	-2.4	-1.7	1.0	8.6	14.0	18.2	19.8	18.5	13.9	8.8	2.8	-0.8	(24)	
	10%	DB	-3.4	-2.2	1.9	11.9	21.1	24.6	26.3	23.1	17.0	9.9	2.1	-1.5	(25)	
		MCWB	-4.2	-3.3	0.2	6.7	13.1	17.4	18.9	17.5	13.0	7.6	1.2	-2.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	0.9	4.8	13.1	18.0	21.6	22.3	21.7	17.7	12.3	6.7	1.2	(27)	
		MCDB	1.5	1.7	8.8	22.2	25.1	28.4	28.9	28.8	23.0	16.0	7.9	1.6	(28)	
	2%	WB	-1.0	-0.3	2.6	10.8	16.1	20.1	21.2	20.4	15.7	10.5	4.7	0.2	(29)	
		MCDB	-0.2	0.7	4.7	17.6	23.4	26.5	28.1	26.8	20.2	14.3	5.5	0.8	(30)	
	5%	WB	-2.4	-1.6	1.5	9.0	14.9	19.0	20.4	19.1	14.5	9.1	2.8	-0.9	(31)	
		MCDB	-1.6	-0.7	3.1	14.6	21.9	25.0	27.0	24.6	18.7	11.4	3.7	-0.1	(32)	
	10%	WB	-4.2	-3.4	0.3	7.3	13.7	17.9	19.5	17.9	13.3	7.7	1.4	-2.3	(33)	
		MCDB	-3.5	-2.3	1.8	11.3	19.8	23.5	25.2	22.2	16.7	10.1	2.0	-1.5	(34)	
(35) Mean Daily Temperature Range	MDBR		8.9	10.6	10.6	10.6	12.4	11.5	11.5	10.4	9.3	6.5	5.9	7.4	(35)	
	5% DB	MCDBR	6.1	7.3	10.2	16.3	17.3	15.8	15.6	15.0	14.0	13.1	5.6	5.2	(36)	
		MCWBR	5.8	6.4	7.5	9.2	8.3	7.2	7.0	7.2	7.4	8.2	4.6	5.2	(37)	
	5% WB	MCDBR	5.9	6.5	8.5	15.4	14.6	14.0	14.0	14.1	11.9	11.2	5.1	5.4	(38)	
		MCWBR	5.7	5.9	6.6	8.9	7.8	6.9	6.7	7.2	6.7	7.6	4.5	5.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.292	0.318	0.374	0.409	0.403	0.397	0.420	0.438	0.383	0.353	0.313	0.290	(40)	
	taud		2.013	2.017	1.927	2.046	2.247	2.328	2.264	2.185	2.308	2.254	2.108	1.979	(41)	
	Ebn at Noon		546	701	750	795	834	845	813	764	758	669	522	428	(42)	
	Edh at Noon		64	97	138	143	125	117	123	125	95	77	58	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.65	1.58	2.92	4.15	5.11	5.44	5.44	4.14	2.54	1.12	0.53	0.37	(44)	
	RadStd		0.09	0.22	0.30	0.52	0.57	0.54	0.58	0.43	0.22	0.22	0.10	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page